

DEFINITION — HOW DATA IS DESCRIBED

API INTERFACES6

**ODCS**  
BITOL @ LF

**OpenAPI**  
LF

**AsyncAPI**  
LF

**GraphQL**  
LF

**gRPC**  
CNCF

**OData**  
OASIS

DATA PRODUCTS4

**ODPS**  
BITOL @ LF

**DPDS**  
ODM

**ODPSpec**  
LF

**DPROD**  
OMG

SCHEMA5

**XML Schema**  
W3C

**JSON Schema**  
IETF

**SQL DDL**  
ISO/IEC

**AVRO Schema**  
ASF

**Protobuf**  
GOOGLE

SEMANTICS6

**OSI**  
OSI INITIATIVE

**RDF/OWL**  
W3C

**DCAT**  
W3C

**SKOS**  
W3C

**SHACL**  
W3C

**JSON-LD**  
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS6

**CSV**  
IETF

**JSON**  
IETF

**XML**  
W3C

**PARQUET**  
ASF

**AVRO**  
ASF

**ORC**  
ASF

OPEN TABLE FORMATS3

**Iceberg**  
ASF

**Delta**  
LF

**Hudi**  
ASF

STORAGE SYSTEMS2

**S3**  
AWS

**HDFS**  
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY3

**JDBC**  
JCP

**ODBC**  
ISO/IEC

**ADBC**  
ASF

INTERCONNECTION3

**HTTP**  
IETF

**FTP / SFTP**  
IETF

**Delta Sharing**  
LF

MESSAGING5

**Kafka**  
ASF

**MQTT**  
OASIS

**CloudEvents**  
CNCF

**AMQP**  
OASIS

**JMS**  
JAKARTA EE

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2

**DataFrame API**  
DATA-APIS

**Apache Arrow**  
ASF

PROCESSING4

**dbt**  
DBT LABS

**Spark**  
ASF

**Pandas**  
NUMFOCUS

**Ibis**  
IBIS

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS4

**Iceberg Catalog**  
ASF

**Unity Catalog**  
LF

**Hive Metastore**  
ASF

**Schema Registry**  
CONFLUENT

LINEAGE1

**OpenLineage**  
LF

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY2

**SQL**  
ISO/IEC

**SPARQL**  
W3C

DATA QUALITY3

**dbt**  
DBT LABS

**Great Expectations**  
GX LABS

**SodaCL**  
SODA

OBSERVABILITY2

**Open Telemetry**  
CNCF

**OORS**  
BITOL @ LF

POLICIES1

**OPA**  
CNCF

AI INTERFACES2

**MCP**  
LF

**A2A**  
LF